The Jäschke–Skogestad temperature conjecture for heat-exchanger networks is false: disproof and corrected theorem

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Abstract. Jäschke and Skogestad (Comput. Chem. Eng. 2014) showed that optimal operation of a heat-exchanger network with stream splits only requires temperature measurements: with an arithmetic-mean temperature-difference model, the marginal heat recovered by a split branch is a sum $\sum_i p_{ij} a_{ij}$ of recursively defined "Jäschke temperatures". They proved this for branches with $M_j \le 2$ exchangers and conjectured it for $M_j \ge 4$ (the case $M_j = 3$ being proved only for constant hot-stream temperatures). We show that the conjectured recursion is false for every $M_j \ge 3$, including the published proof of $M_j = 3$, where a term is lost in a manual simplification; an exact rational counterexample has the paper's controlled variable equal to zero while the true gradient is $-4972391424/1518467647$. We prove the corrected recursion $a_i = c_i(\theta_i + \theta_{i-1} - \sum_{l<i} a_l)$ for every M_j, with a half-page proof based on the affine structure of each exchanger, and show that it explains the discrepancy that the paper attributed to round-off in its own validation case.

1 Source and exact statement

J. Jäschke, S. Skogestad, *Optimal operation of heat exchanger networks with stream split: Only temperature measurements are required*, Computers and Chemical Engineering 70 (2014) 35–49, doi:10.1016/j.compchemeng.2014.03.020 (Manfred Morari special issue). Read in full: the publisher PDF self-archived at `https://skoge.folk.ntnu.no/publications/2014/jaschke-hen-paper-cce-morari-issue/impress.pdf`, plus the older preprint `.../old-version/HenCCE.pdf`. Page numbers below are journal page numbers.

Model (Section 3.1, p. 37–38). N parallel branches split from a cold feed (temperature T_0, heat-capacity flow rate w_0); branch j carries $w_j = F_j c_{p0}$, $\sum_j w_j = w_0$, and M_j heat exchangers in series. For exchanger i on branch j: cold outlet $T_{i,j}$ (cold inlet $T_{i-1,j}$, $T_{0,j} = T_0$), hot inlet $T_{hi,j}$, hot outlet $T^{out}_{hi,j}$, hot heat-capacity flow rate $w_{hi,j}$, conductance $UA_{i,j}$:

$$Q_{i,j} = w_j(T_{i,j} - T_{i-1,j}) \quad (9), \qquad Q_{i,j} = w_{hi,j}(T_{hi,j} - T^{out}_{hi,j}) \quad (10),$$

$$Q_{i,j} = UA_{i,j}\,\frac{(T_{hi,j} - T_{i,j}) + (T^{out}_{hi,j} - T_{i-1,j})}{2} \quad (12,\ \text{Assumption 3, AMTD}).$$

Assumptions 1–2: single phase, constant heat capacities. Cost (14): $J = -\sum_j \sum_i p_{i,j} Q_{i,j}$. Shifted temperatures (15): $\theta = T - T_0$.

Theorem 2 (p. 39), verbatim. "*Under Assumptions 1–3, and equal prices $p_{i,j} = 1$ in the cost function (14), the marginal costs for each branch $j = 1 \dots N$ can be expressed as*

$$\frac{\partial J}{\partial w_j} = T_{J,j} \quad (16)$$

where $T_{J,j}$ is the Jäschke Temperature on branch j, defined as $T_{J,j} = \sum_{i=1}^{M_j} a_{i,j}$ (17), with the parameter $a_{i,j}$ defined recursively as

$$a_{i,j} = \frac{(\theta_{i,j} - \theta_{i-1,j})(\theta_{i,j} + \theta_{i-1,j} - a_{i-1,j})}{\theta_{hi,j} - \theta_{i-1,j}}, \qquad a_{0,j} = 0. \quad (18)"$$

Theorem 3 (p. 39) is the same with arbitrary prices: $\partial J/\partial w_j = T^e_{J,j} = \sum_i p_{i,j} a_{i,j}$ (19)–(20), same $a_{i,j}$.

The open statement, verbatim. - p. 39, after Theorem 2: "*In Appendix A we provide a proof of Theorem 2 for $M_j = 1$ and $M_j = 2$. Moreover, we prove the case $M_j = 3$ for heat exchangers with constant hot stream temperatures (this corresponds to very large hot stream heat capacities). For $M_j \geq 4$ we conjecture that the Theorem is true.*" The same sentence is repeated after Theorem 3. - p. 47, end of A.3: "*Although this expression was derived under the assumption of a constant hot stream temperature $T_{hi,j} = T^{out}_{hi,j}$, we conjecture that it is true for the case with a finite heat capacity (non-constant hot streams).*" - p. 47, A.4 "Conjecture for the general case ($M \geq 4$)": "*Unfortunately it was computationally not feasible to perform the elimination for the case with $M_j > 3$. From the recurring pattern, that coefficients of the previous prices appear in a fixed structure in the coefficients of the next price, we conjecture the general formula for the case with more than three heat exchangers on a line as*" (A.14)–(A.16), identical to (18)–(20). - p. 45, Conclusions: "*For the case with 1–3 heat exchangers per line, we have proven [...] that the Jäschke Temperatures are equal to the marginal costs. The more general case with more than 3 heat exchangers is conjectured.*" - p. 49, Appendix B: "*In conclusion, the numerical results for this case study strongly suggest that the conjecture is true.*"

The recursion is unambiguous: (A.13) and (B.1) write it out for $M_j = 3, 4$, and each new term subtracts only the previous $a_{i-1,j}$.

The scout's summary (rapports/scout_E_mining.md §2) is accurate, except that the conjecture concerns Theorems 2 **and** 3 and also covers $M_j = 3$ with finite hot heat capacities.

2 Results

Fix one branch and drop the index j. Write w for its cold heat-capacity flow rate, $\theta_0 = 0$, and $\Delta_i = \theta_i - \theta_{i-1}$. For each exchanger define

$$k_i = \frac{2}{UA_i} + \frac{1}{w_{h,i}} \quad (1/\infty := 0), \qquad c_i = \frac{\theta_i - \theta_{i-1}}{\theta_{h,i} - \theta_{i-1}}$$

(the paper's "efficiency", eq. 27).

Theorem A (corrected Jäschke temperature; all $M \geq 1$). Under Assumptions 1–3, for every exchanger i of the branch,

$$\frac{\partial Q_i}{\partial w} = \frac{a_i}{2}, \qquad a_i = c_i\Big(\theta_i + \theta_{i-1} - \sum_{l<i} a_l\Big),$$

where $\partial/\partial w$ holds T_0, the hot inlet temperatures, the $w_{h,i}$ and the UA_i fixed. Hence

$$\frac{\partial J_j}{\partial w_j} = -\tfrac{1}{2}\sum_{i=1}^{M_j} p_{i,j}\, a_{i,j}, \qquad \text{and for } p_{i,j}\equiv 1:\ \frac{\partial J_j}{\partial w_j} = -\tfrac{1}{2}\, T^*_{J,j},$$

where $T^*_{J,j} = A_{M_j}$ is given by the affine recursion

$$A_0 = 0, \qquad A_i = (1-c_i)\, A_{i-1} + c_i\,(\theta_i + \theta_{i-1}).$$

This is the same form as the cold-stream recursion $\theta_i = (1-c_i)\theta_{i-1} + c_i\theta_{h,i}$. The formula uses only the cold temperatures $\theta_0, \dots, \theta_M$ and the hot inlet temperatures $\theta_{h,i}$. Because the factor $-\frac{1}{2}$ is common to all branches, equal marginal costs (Theorem 1 of the paper) is **exactly** equivalent to $T^{e*}_{J,1} = \cdots = T^{e*}_{J,N}$.

Corollary B (the conjecture is false). Let a_i^P denote the paper's recursion (18), and $d_i = a_i^P - a_i$. Then $d_1 = d_2 = 0$ and $d_i = c_i\,(A_{i-2} - d_{i-1})$ for $i \geq 3$. In particular $d_3 = c_3\, a_1 = c_3\,\theta_1^2/\theta_{h,1}$, which is nonzero whenever exchangers 1 and 3 both transfer heat. Consequently: 1. The paper's formula is correct for $M_j \leq 2$ up to the harmless common factor $-\frac{1}{2}$ (Section 4.4). So Examples 1–2 and Case studies 1–3 of the paper are unaffected. 2. **Theorem 3 (prices) is false for every $M_j \geq 3$, even up to any normalisation constant.** Suppose $\sum_i p_i a_i^P = \lambda \sum_i p_i a_i$ for all prices. Then $a_i^P = \lambda a_i$ for every i. The case $i = 1$ ($a_1^P = a_1 \neq 0$) forces $\lambda = 1$, but $a_3^P = a_3 + c_3 a_1 \neq a_3$. 3. **Theorem 2 ($p \equiv 1$) is false for every $M_j \geq 3$.** Section 3.4 gives an exact rational counterexample for $M_1 = 3, M_2 = 1$: the paper's controlled variable is exactly 0 while the true gradient is not. This is independent of any normalisation. For $M \geq 4$, append to branch 1 of that example $M-3$ exchangers with conductance ε and hot inlet temperatures different from θ_3. As $\varepsilon \to 0^+$, $c_i = 2/(1 + w(2/\varepsilon + 1/w_{h,i})) \to 0$ for $i \geq 4$, so $d_i \to 0$ for $i \geq 4$. Hence $T_J^P - T_J^* = \sum_i d_i \to c_3 a_1 \neq 0$, and the difference is nonzero for small $\varepsilon > 0$. The computer-algebra checks for $M = 3, \dots, 8$ and the random tests for $M \leq 8$ find the difference to be a nonzero rational function / nonzero in every draw. 4. The case the paper reports as **proved**, $M_j = 3$ with constant hot temperatures, is also wrong as printed. The first line of the paper's (A.12), the raw Maple coefficient of $p_{3,1}$, equals our a_3 exactly. The "simplification" in the next lines of (A.12) silently drops $-a_{1,1}$ (checked symbolically, claim S4). The paper's elimination was right; the hand simplification and the extrapolated pattern (A.15) are wrong.

Proposition C (any exchanger law that is linear in temperatures; LMTD). Suppose every exchanger satisfies $\Delta_i = \varepsilon_i\,(\theta_{h,i} - \theta_{i-1})$, where the cold-side effectiveness ε_i depends on w and on the exchanger parameters but not on the temperatures. This holds for AMTD and for LMTD/ε–NTU models of any flow arrangement. Put $\varphi_i = w\,\partial\varepsilon_i/\partial w$. Then

$$\frac{\partial Q_i}{\partial w} = \Big(1 + \frac{\varphi_i}{\varepsilon_i}\Big)\Delta_i + \varepsilon_i\Big(\theta_{i-1} - \sum_{l<i}\frac{\partial Q_l}{\partial w}\Big).$$

- For AMTD, $\varphi = -\varepsilon(2-\varepsilon)/2$, so $1 + \varphi/\varepsilon = \varepsilon/2$; this gives Theorem A. - For a counter-current exchanger with the logarithmic mean (LMTD), $\varepsilon = (1-E)/(1-RE)$ with $E = e^{-N(1-R)}$, $N = UA/w$, $R = w/w_h$, and $\varphi = -N\,\partial_N\varepsilon + R\,\partial_R\varepsilon$. Now ε and $R = (\theta_h - \theta_h^{out})/\Delta$ are fixed by the four terminal temperatures, and $N = \ln((1-R\varepsilon)/(1-\varepsilon))/(1-R)$. So the exact LMTD marginal cost is again a function of temperatures only, but it needs the **hot outlet** temperatures as well. - For condensing hot streams ($R = 0$), $\varphi = (1-\varepsilon)\ln(1-\varepsilon)$ and no hot outlet temperature is needed. -

Under AMTD the R-dependence cancels exactly: $\varepsilon = 2/(1 + 2/N + R)$ depends on w only through wk_i. This is why the AMTD formula needs no heat capacities and no hot outlet temperatures. - Applied to an LMTD plant, the (corrected) AMTD formula is only an approximation. In our random tests its relative error on the branch marginal cost has median $\approx 9\%$, with large errors for very unbalanced exchangers, consistent with the paper's own discussion.

3 Proofs

3.1 3.1 Lemma 1 (each exchanger is an affine map of the cold temperature)

For given θ_{i-1}, the three equations (9), (10), (12) have the unique solution

$$\theta_i = \theta_{i-1} + c_i(\theta_{h,i} - \theta_{i-1}), \qquad c_i = \frac{2}{1 + wk_i}\,, \qquad Q_i = w\,\Delta_i, \qquad \theta_{h,i}^{out} = \theta_{h,i} - Q_i/w_{h,i}.$$

For $w_{h,i} = \infty$, take $\theta_{h,i}^{out} = \theta_{h,i}$ and drop (10).

Proof. Substitute $\theta_{h,i}^{out} = \theta_{h,i} - Q_i/w_{h,i}$ and $\theta_i = \theta_{i-1} + Q_i/w$ into (12):

$$Q_i = UA_i\Big(\theta_{h,i} - \theta_{i-1} - \frac{Q_i}{2w} - \frac{Q_i}{2w_{h,i}}\Big) \iff Q_i\Big(\frac{1}{UA_i} + \frac{1}{2w} + \frac{1}{2w_{h,i}}\Big) = \theta_{h,i} - \theta_{i-1}.$$

The bracket is positive, so Q_i is unique. Dividing by w gives $\Delta_i = 2(\theta_{h,i} - \theta_{i-1})/(1 + wk_i)$. □

Remarks. - c_i is the temperature ratio c_i of Section 2 whenever $\theta_{h,i} \neq \theta_{i-1}$. - $(UA_i, w_{h,i})$ enter only through k_i. So, on the cold side, an exchanger with finite $w_{h,i}$ is identical to a constant-hot-temperature exchanger with $UA' = (1/UA_i + 1/(2w_{h,i}))^{-1}$. This makes the paper's reduction to constant hot temperatures in A.3 rigorous, and explains why the conjectured extension "to finite heat capacity" is automatic. - The paper's (A.5), $\theta_j = \theta_{hj}/(1 - w_j(1/w_{hj} + 2/UA_j))$, has a sign error and a missing factor 2. Section 4.4 explains why this does not affect the controlled variable.

3.2 3.2 Lemma 2

$$w\,\frac{\partial c_i}{\partial w} = -\,\frac{c_i(2 - c_i)}{2}\,.$$

Proof. $\partial c_i/\partial w = -2k_i/(1 + wk_i)^2$, and $wk_i = (2 - c_i)/c_i$, so $w\,\partial c_i/\partial w = -2\,\frac{2-c_i}{c_i}\,\frac{c_i^2}{4} = -\,\frac{c_i(2-c_i)}{2}$. □

3.3 3.3 Proof of Proposition C (and so of Theorem A)

Let $D_i = w\,\partial\theta_i/\partial w$, with $D_0 = 0$ because T_0 is fixed.

Step 1 (cumulative balance). Summing (9) over $l \le i$ gives $\sum_{l\le i} Q_l = w\theta_i$. Differentiating,

$$\sum_{l\le i} \frac{\partial Q_l}{\partial w} = \theta_i + D_i \qquad (i = 0, \ldots, M). \tag{$*$}$$

Step 2 (chain rule through the affine maps). From $\theta_i = \theta_{i-1} + \varepsilon_i(\theta_{h,i} - \theta_{i-1})$ with $\theta_{h,i}$ fixed,

$$D_i = (1 - \varepsilon_i)D_{i-1} + \varphi_i(\theta_{h,i} - \theta_{i-1}) = (1 - \varepsilon_i)D_{i-1} + \frac{\varphi_i}{\varepsilon_i}\,\Delta_i.$$

For $\theta_{h,i} = \theta_{i-1}$, read $\frac{\varphi_i}{\varepsilon_i}\,\Delta_i$ as $\varphi_i(\theta_{h,i} - \theta_{i-1}) = 0$.

Step 3. Taking the difference of $(*)$ at i and $i-1$,

$$\frac{\partial Q_i}{\partial w} = \Delta_i + D_i - D_{i-1} = \Big(1 + \frac{\varphi_i}{\varepsilon_i}\Big)\Delta_i - \varepsilon_i D_{i-1}.$$

Eliminating D_{i-1} with $(*)$ at $i-1$, namely $D_{i-1} = \sum_{l<i} \partial Q_l/\partial w - \theta_{i-1}$, gives Proposition C.

AMTD. With $\varepsilon_i = c_i$, Lemma 2 gives $1 + \varphi_i/c_i = c_i/2$, so

$$\frac{\partial Q_i}{\partial w} = \frac{c_i}{2}\Big(\Delta_i + 2\theta_{i-1} - 2\sum_{l<i}\frac{\partial Q_l}{\partial w}\Big) = \frac{c_i}{2}\Big(\theta_i + \theta_{i-1} - \sum_{l<i} 2\,\frac{\partial Q_l}{\partial w}\Big).$$

Setting $a_i := 2\,\partial Q_i/\partial w$ gives exactly the recursion of Theorem A, which determines the a_i uniquely from a_1 on. The statements about J_j follow from $J_j = -\sum_i p_{i,j} Q_{i,j}$. Since $A_i = \sum_{l\le i} a_l$, we get $A_i = A_{i-1} + c_i(\theta_i + \theta_{i-1} - A_{i-1})$, which is the affine form. □

Corollary B, the recursion for d_i. $a_i^P - a_i = c_i\big(\sum_{l<i} a_l - a_{i-1}^P\big) = c_i(A_{i-2} + a_{i-1} - a_{i-1}^P) = c_i(A_{i-2} - d_{i-1})$, with $d_1 = 0$ (both equal $c_1\theta_1$) and $d_2 = c_2(A_0 - d_1) = 0$.

3.4 3.4 Exact counterexample ($M_1 = 3$, $M_2 = 1$, prices 1)

Produced and checked by `scripts/counterexample_exact.py`; data in `certificates/counterexample.json`. All values are exact rationals.

Data. - $T_0 = 20$ °C, $w_0 = 100$ kW/K, split $w_1 = w_2 = 50$ kW/K. - Branch 1: $UA = (56, 34, 28)$ kW/K, $w_h = (88, 51, 78)$ kW/K, $\theta_h = (48, 52, 60)$ K, i.e. hot inlets at 68, 72 and 80 °C. - Branch 2: $UA_{1,2} = 200$, $w_{h1,2} = 50$, $\theta_{h1,2} = 13184889712830714450/136131266865364681 \approx 96.854$ K. This value is chosen as $T_{J,1}^P/c^2$ with $c = 4/5$.

What the operating point satisfies. - It is physical: in every exchanger the cold stream is heated, the cold outlet stays below the hot inlet, the hot outlet stays above the cold inlet, and $0 < c_i < 1$. The largest ratio of terminal temperature differences is 1.64, so AMTD is a reasonable approximation of LMTD here. - Branch-1 temperatures: $\theta_1 \approx 28.623$, $\theta_2 \approx 38.123$, $\theta_3 \approx 46.517$. - The paper's controlled variable vanishes exactly: $T_{J,1} - T_{J,2} = 0$, with $T_{J,1} = T_{J,2} \approx 61.98671$. - The exact gradient of the recovered heat, computed by exact forward-mode differentiation of the raw model (no formula used), is

$$\frac{dQ_{tot}}{dw_1}\Big|_{w_2 = w_0 - w_1} = -\frac{4972391424}{1518467647} \approx -3.2746 \neq 0.$$

This equals $(T_{J,1}^* - T_{J,2}^*)/2 = -c_3 a_1/2$, with $T_{J,1}^* \approx 55.43749$. - The split $w_1' = 99/2$ recovers strictly more heat: by about 1.47 kW, exactly.

So the point where the paper's controller settles is not optimal. Numerically (illustration only), the true optimum is $w_1^\star \approx 47.504$ ($u^\star \approx 0.4750$ instead of 0.5), and the loss is ≈ 4.08 kW (0.066 % of 6204 kW). The loss is small because the optimum is flat; the split error is about 5 % of w_0.

3.5 3.5 The paper's own validation case (Appendix B)

`scripts/reproduce_appendixB.py` uses Table B.6: $M_1 = 4$, $M_2 = 1$, prices 1, 1.2, 1.3, 1.5 / 1.5. All functions are exact rational functions of $u = w_1/w_0$, and real roots are isolated exactly. - (B1) The corrected controlled variable equals $-\frac{2}{w_0}\,dJ/du$ identically.

- (B2) dJ/du has a unique root in $(0,1)$: $u_{opt} \in [0.7141194663, 0.7141194665]$, with $J = -3938.8167$ and $T_{end} = 158.77375$ °C. This matches the paper's "Optimized" column (0.7141, −3.9388, 158.7738). - (B3) The paper's controlled variable (B.1), with the misprints of the printed (B.1) corrected, vanishes at $u_P \in [0.7167062190, 0.7167062200] \neq u_{opt}$. Its only other root in $(0,1)$, $u \approx 0.0028$, lies in the unphysical AMTD region $c_{1,1} > 1$.

The paper reports $u = 0.7152$ for "Equal Jäschke Temp.". We could not reproduce this number from (B.1), with or without the printing typos ($\theta^2_{h1,1}$ for $\theta^2_{1,1}$, and $\theta_{h3,1} - \theta_{2,1}$ for $\theta_{h2,1} - \theta_{1,1}$ inside the $p_{3,1}$ term). Either way, the split discrepancy that the paper attributes to "flatness of the optimum and numerical round-off errors" is real and is explained by Corollary B.

4 Verification summary, scope and limitations

4.1 4.1 Computer algebra (`scripts/verify_symbolic.py`, python-flint)

- The raw model of Section 1 is solved by Cramer's rule in $\mathbb{Q}(w, UA_i, w_{h,i}, \theta_{h,i})$ with fully generic symbols, and $\partial Q_i/\partial w$ is computed symbolically.
- **S1:** it equals $a_i/2$ (corrected recursion) as an identity of rational functions for every $i \leq M$. This is checked for $M = 1, \ldots, 5$ with finite $w_{h,i}$, all symbols generic (up to 16 indeterminates), and for $M = 1, \ldots, 8$ with $w_{h,i} = \infty$.
- **L1:** a single exchanger with a generic inlet temperature gives the same outlet as a constant-hot-temperature exchanger with $UA' = (1/UA + 1/(2w_h))^{-1}$. This is the symbolic form of the remark after Lemma 1. Because UA' does not depend on w, the $w_{h,i} = \infty$ checks carry over to finite $w_{h,i}$ for all $M \leq 8$.
- The raw finite-w_h model at $M = 6$ (19 indeterminates) exceeded the 4 GB memory cap after passing S0 and S1 for $i \leq 5$ (`certificates/verify_symbolic_M6_raw_partial.log`).
- **S2, S3:** the paper's recursion is identical for $i \leq 2$, which is a symbolic identity. It differs for every $i \geq 3$, its branch total differs from $2\,\partial \sum_i Q_i/\partial w$, and $d_3 = c_3 a_1$. Non-vanishing is certified by exact evaluation at a rational point where no denominator vanishes.
- **S4:** the paper's printed Maple numerator in (A.12) equals the corrected a_3.
- These checks are redundant with the proof of Section 3, which does not rely on them.

4.2 4.2 Numerical evidence (`scripts/random_fd_tests.py`)

This is independent evidence, not the proof. - 60-digit central finite differences on random data, $M = 1..8$, with random prices: the corrected formula agrees to relative error $<$ 1e−30. - The paper's T_J is off by at least 0.28 % in every draw with $M \geq 3$. - The LMTD form of Proposition C agrees to $<$ 1e−25.

4.3 4.3 Domain of validity

- Theorem A is an identity of rational functions. It holds wherever the model is defined ($1 + wk_i \neq 0$, automatically true for positive parameters). There is no sign restriction: it also holds for exchangers that cool the branch, and in the unphysical AMTD regime $c_i > 1$ (small w).
- The temperature-only expression of c_i needs $\theta_{h,i} \neq \theta_{i-1}$. At $\theta_{h,i} = \theta_{i-1}$ the exchanger transfers no heat, yet $\partial Q_i/\partial w = a_i/2 = \frac{c_i}{2}(2\theta_{i-1} - A_{i-1})$ is generally nonzero, with $c_i = 2/(1 + wk_i)$ no longer measurable. So the paper's regularisation

in §6.6.1 (setting $a_{i,j} = 0$ near this singularity) is itself only approximate.

- The model assumes that the hot-side data $(\theta_{h,i}, w_{h,i}, UA_i)$ do not depend on the split, i.e. no stream crossovers between branches. This is the same restriction as in the paper (§6.5).
- With LMTD and finite hot heat capacities, φ_i/ε_i depends on R_i as well as on ε_i. We have not proved that no exact formula in cold temperatures and hot inlet temperatures alone exists, but Proposition C gives an exact one that also uses the hot outlet temperatures. In the paper's §5 simulations the AMTD-based controlled variable is a (good) heuristic for LMTD plants. For $M_j \geq 3$ the corrected formula should be used.

4.4 4.4 The factor $-\frac{1}{2}$

- Literally, (16) and (19) are off by the factor $-\frac{1}{2}$ even for $M_j = 1$. For a single exchanger the correct relation is $\partial J_j/\partial w_j = -\theta_{1,j}^2/(2\theta_{h1,j})$.
- The sign comes from $J = -\sum pQ$, written as "$\partial J_j/\partial w_j = p_j\, \partial Q_j/\partial w_j$" in (A.6). The 2 comes from the mis-solved (A.5).
- The factor is the same for every branch, so the optimality condition "equal Jäschke temperatures" is unaffected. We regard this as a normalisation slip, not as part of the conjecture.

4.5 4.5 Points a referee should check first

1. Lemma 1: the solution of the AMTD exchanger equations and the factor 2.
2. Step 1, $\sum_{l \leq i} Q_l = w\theta_i$, which is the source of the cumulative sum.
3. The comparison with the paper's (A.12): the transcription of the 12-term numerator from the rendered page 47 (claim S4).
4. That the paper's recursion really subtracts only a_{i-1}: (18), (A.13), (A.15) and (B.1) all write it that way.